Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

Learning goal: I can find the sample space of an experiment. (circle your level after scoring)

4	I can explain why a repeated color or letter counts as one outcome, and decide whether two experiments have the same sample space.
3	I can list or count every outcome in the sample space, with no repeats or misses.
2	I can list some of the outcomes of a simple experiment.
1	I need help to start this benchmark.

1. Each letter of the word DIFFERENCE is written on its own card, one card for each letter, and the cards are placed in a jar. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 8

B. 7

C. 10

D. 6

2. A set of cards shows these colored shapes, one on each card: red circle, yellow star, yellow triangle, yellow triangle, yellow star, yellow triangle. One card is drawn at random, and the colored shape on it is recorded.

List the sample space for this experiment.

Turn over →

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____

3. Experiment 1: A spinner has 5 equal sections colored red, green, orange, purple and red. It is spun once and the color it lands on is recorded.
Experiment 2: A bag holds 1 orange, 3 green and 2 red marbles. One marble is drawn at random and its color is recorded.

Which statement about the sample spaces of the two experiments is true?

A. The sample space for the spinner has one outcome that the sample space for the bag does not have.

B. The sample space for the spinner has 5 outcomes, and the sample space for the bag has 6 outcomes.

C. The two experiments have the same sample space, {orange, green, red, purple}.

D. The sample space for the bag has one outcome that the sample space for the spinner does not have.

4. A fair 8-sided die has these numbers on its faces: 5, 5, 1, 8, 9, 8, 9, 5. The die is rolled once and the number on the top face is recorded.

List the sample space for this experiment.

5. In each experiment below, the color is recorded.

Select **all** the experiments whose sample space is {purple, green, blue}.

Select **all** the correct answers.

A. ☐ Rolling a fair 6-sided die whose faces are blue, yellow, green, purple, green, purple

B. ☐ Spinning a spinner whose 6 equal sections are green, blue, purple, green, purple, blue

C. ☐ Drawing a marble from a bag that holds 3 purple, 5 green and 4 blue marbles

D. ☐ Spinning a spinner whose 4 equal sections are green, purple, green, purple

E. ☐ Drawing a marble from a bag that holds 3 purple, 5 green, 4 blue and 4 yellow marbles

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.2.1 · MA.7.DP.2.1 · Sample space of a simple experiment

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **B. 7**

DIFFERENCE has 10 letters, so there are 10 cards, but F is on 2 cards and E is on 3 cards.
The outcome is the letter that is recorded, and a letter that is on more than one card is still one outcome.
The sample space is $\{D, I, F, E, R, N, C\}$, which has 7 outcomes, not 10.

2. Regular **{red circle, yellow star, yellow triangle}**

There are 6 cards, but yellow star is on 2 cards and yellow triangle is on 3 cards.
The outcome is the colored shape, and cards that show the same colored shape give the same outcome, so each different colored shape is listed once.
The sample space is {red circle, yellow star, yellow triangle}.

3. Regular **A. The sample space for the spinner has one outcome that the sample space for the bag does not have.**

Each outcome is a color, and a color is listed once no matter how many sections or marbles have it.
Spinner: {orange, green, red, purple}. Bag: {orange, green, red}.
The spinner can land on purple, and the bag has no purple marble, so the sample space for the spinner has one more outcome.

4. Regular $\{1, 5, 8, 9\}$

The die has 8 faces, but some faces show the same number.
Each different number is one outcome, so a number that is on more than one face is listed only once.
The sample space is $\{1, 5, 8, 9\}$.

5. Rigor **B. Spinning a spinner whose 6 equal sections are green, blue, purple, green, purple, blue; C. Drawing a marble from a bag that holds 3 purple, 5 green and 4 blue marbles**

List the colors each experiment can give, each color once:
Drawing a marble from a bag that holds 3 purple, 5 green and 4 blue marbles: {purple, green, blue}.
Spinning a spinner whose 6 equal sections are green, blue, purple, green, purple, blue: {green, blue, purple}.
Rolling a fair 6-sided die whose faces are blue, yellow, green, purple, green, purple: {blue, yellow, green, purple}.
Drawing a marble from a bag that holds 3 purple, 5 green, 4 blue and 4 yellow marbles: {purple, green, blue, yellow}.
Spinning a spinner whose 4 equal sections are green, purple, green, purple: {green, purple}.
Only the experiments whose list is exactly {purple, green, blue} have this sample space; a color on several sections, faces or marbles is still one outcome.
Scoring: 1 point only if every correct box is checked and no other box is; anything else earns 0.

Re-check strip (3 items, C-DP.2.1-R)

1. Regular **B. 4**

DEGREE has 6 letters, so there are 6 cards, but E is on 3 cards.
The outcome is the letter that is recorded, and a letter that is on more than one card is still one outcome.
The sample space is $\{D, E, G, R\}$, which has 4 outcomes, not 6.

2. Regular **A. {blue, red, purple, green}**

The outcome that is recorded is a color, so the sample space lists each color once: {blue, red, purple, green}.
The numbers of sections tell how many sections have each color, and the numbers 1 to 6 would name sections, not colors.

3. Regular $\{1, 2, 8\}$

There are 7 tiles, but 1 is on 2 tiles, 2 is on 2 tiles and 8 is on 3 tiles.
The outcome is the number, and a number that is on more than one tile is still one outcome, so each different number is listed once.
The sample space is $\{1, 2, 8\}$.

Comparison strip (2 items, C-DP.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **B. 6**

ADDITION has 8 letters, so there are 8 cards, but D is on 2 cards and I is on 2 cards.
The outcome is the letter that is recorded, and a letter that is on more than one card is still one outcome.
The sample space is $\{A, D, I, T, O, N\}$, which has 6 outcomes, not 8.

2. Regular $\{1, 6, 9\}$

The die has 6 faces, but some faces show the same number.
Each different number is one outcome, so a number that is on more than one face is listed only once.
The sample space is $\{1, 6, 9\}$.

Reteach plan — C-DP.2.1

MA.7.DP.2.1 · Sample space of a simple experiment

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.1-R

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

1. Each letter of the word DEGREE is written on its own card, one card for each letter, and the cards are placed in a hat. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 6

B. 4

C. 5

D. 3

2. A spinner has 6 equal sections: 1 is blue, 1 is red, 1 is purple and 3 are green. The spinner is spun once and the color it lands on is recorded.

The sample space for this experiment is ____.

Choose the option that correctly completes the statement.

A. {blue, red, purple, green}

B. {blue, red, purple}

C. $\{1, 1, 1, 3\}$

D. $\{1, 2, 3, 4, 5, 6\}$

3. The tiles in a bag are numbered 8, 1, 1, 8, 2, 8, 2. One tile is drawn at random and its number is recorded.

List the sample space for this experiment.

Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2) C-DP.2.1-X

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

1. Each letter of the word ADDITION is written on its own card, one card for each letter, and the cards are placed in a jar. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 8 **B.** 6

C. 5 **D.** 7

2. A fair 6-sided die has these numbers on its faces: 6, 9, 9, 9, 9, 1. The die is rolled once and the number on the top face is recorded.

List the sample space for this experiment.

Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2) C-DP.2.1-X

Benchmark: **MA.7.DP.2.1** · Sample space of a simple experiment Student code: _____ Calculator: yes

1. Each letter of the word ADDITION is written on its own card, one card for each letter, and the cards are placed in a jar. One card is drawn at random and its letter is recorded.

How many outcomes are in the sample space for this experiment?

A. 8 **B.** 6

C. 5 **D.** 7

2. A fair 6-sided die has these numbers on its faces: 6, 9, 9, 9, 9, 1. The die is rolled once and the number on the top face is recorded.

List the sample space for this experiment.

Score: ____ **/ 2** → **copy to your progress strip**